

THE CUSTOMER'S PAIN

You see a charge you never made. Then you typically wait 37 hours to hear back.

A customer who reports a charge they do not recognize waits a **median of 37 hours** for the bank's first response. One in ten waits **58 hours or more**. Until then, the charge sits on their account with no answer.

37 h

median wait for a first response to "Cargo no reconocido" (unrecognized charge)

MEASURED

p90 58 h

n = 7,567 of 12,297

Why disputes: a dispute can be checked in code against the customer's own ledger and decided by an explicit policy.

DESIGN ARGUMENT

Source: docs/analysis/demand-report.md (finding 3: median 37.0 h, p90 58.0 h over the 7,567 of 12,297 complaints with a recorded first response). Measured on the challenge's synthetic LATAM Bank dataset, not a live bank.

A dispute agent that knows when to act, when to ask, and when to step aside.

1 Resolves, with proof

RESUELTO

✓

Identidad verificada
Sesión de demostración activa

✓

Transacción localizada
Cargo identificado: Uber · COP 38.500 · 14 de jun de 2026

✓

Política evaluada
Resolución automática aplicada

The record backs the claim, so a provisional credit lands in the same chat.

2 Asks when it is unclear

Veo que tenemos 2 cargos que coinciden con lo que usted mencionó. Por favor, toque el que no reconoce o seleccione "No está en la lista" si no aparece allí.

Taxi Seguro
COP 27.000
16 de jun de 2026 · Transporte

Taxi Seguro
COP 27.000
15 de jun de 2026 · Transporte

No está en la lista

Escriba su mensaje...

Two matching charges: the customer picks from their own. It never guesses.

3 Steps aside, file ready

HECHOS VERIFICADOS **Del registro**

Transacción

SYN-DEMO-ONLINE

Comercio

Tienda Online Global

Monto

COP 689.000

Fecha

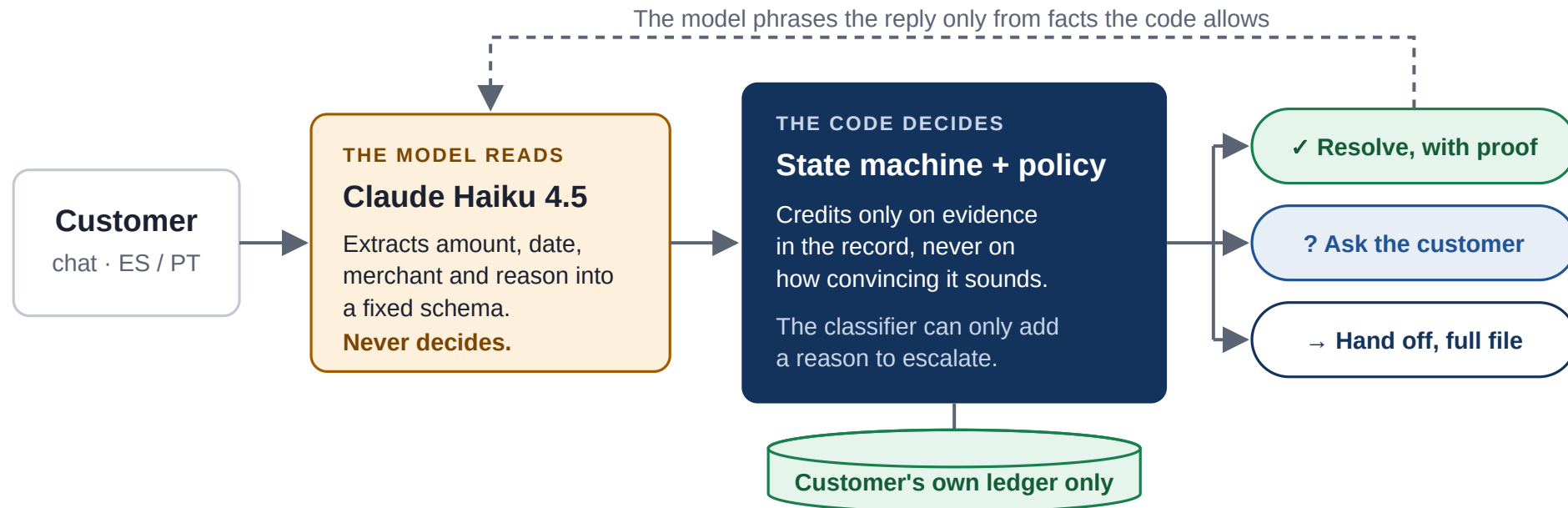
12 de jun de 2026

Canal

Web

Fails the policy: a person gets verified facts, the customer's account and open questions.

The model reads. The code decides.



The LLM only extracts and phrases
No refund comes from the model.

Policy and permissions live in code
No lookup can reach another customer's ledger.

Verified facts only
A persuasive claim the record contradicts still goes to a person.

Design argument, enforced in code and tested: an exhaustive sweep over every dispute reason and all 128 combinations of the other conditions shows the classifier never causes a credit (tests/test_policy_not_overridden.py) · 990 automated tests. Sources: README "Architecture at a glance", docs/architecture-decisions.md AD-3, AD-5, AD-13.

Built on the full challenge dataset, with every shortcut explained.

ETL with contracts

CONTRACTS

9 tables, one schema contract each

QUALITY CHECKS

Row counts, duplicate keys, null rates, foreign keys, and a lineage manifest

FOREIGN KEYS MEASURED

✓ **0 orphaned rows**

Why a subset

TRANSACTIONS MEASURED

130,690 of 5M rows: a 30-day window, May 18 to Jun 17, 2026

COMPLAINTS MEASURED

67,095, in full

WHY DESIGN ARGUMENT

No S3 at runtime (privacy), reproducible, free-plan deploy; a dispute only needs the recent ledger

Priority classifier

HELD-OUT TEST

Chronological split: **11,543** train, **2,037** test

MACRO-F1 MEASURED

Baseline 0.166

Model 0.245

ROLE

Can add a reason to escalate, **never approve a credit**

The full dataset where it matters; a small fixture where privacy and reproducibility matter more.

Sources: etl/schema_contract.py, data/lineage_manifest.json, data/extraction_manifest.json, data/classifier_eval_report.json (baseline 0.1662, model 0.2448, +0.0786: a modest gain, reported as such). The live app reads only a fixture: 1 dataset customer and 8 labeled synthetic charges.

40 scripted attacks and edge cases. Zero unsafe outcomes.

0 / 40

unsafe outcomes SIMULATED

3.2 s median to resolve, real Claude Haiku 4.5 MEASURED

0.05-0.14 s for a button tap, no model call MEASURED

SAME 40 CASES SIMULATED	CORRECT RESOLUTIONS OF 6	UNSAFE CREDITS OF 34	CORRECT TRANSFERS OF 30
Our hybrid evidence check in code	6	0	30
Always escalate the safety anchor	0	0	30
No evidence check the ablation	6	4	26

Escalating everything is safe but helps no one. Dropping the evidence check pays 4 credits a person should have reviewed. The hybrid keeps both.

Read with its limits: a constructed offline suite with a mocked model, written by the policy's author. It is not a held-out workload; the only held-out evaluation is the classifier's chronological split.

Sources: data/eval_report.json (unsafe_outcomes, system_comparison); real-model latency from manual runs on 2026-09-30 (resolving turn 1.3-6.6 s over 8 ES/PT runs), README "Evaluation results".

ROUTE TO PRODUCTION

From demo to the bank's front line.

STEP 1

Real identity

The bank's identity verification replaces the demo login.

STEP 2

Managed database

Replicated storage across several instances, not local SQLite.

STEP 3

Monitoring

The structured, correlated event log feeds alerting.

STEP 4

Real traffic

Policy thresholds and the live model validated on held-out cases.

LIVE DEMO

factored-hackaton-latest.onrender.com

CODE

github.com/jAgusGelos/factored-hackathon-2026-jagusgelos

An answer in seconds, not 37 hours. Every refund backed by evidence. Every hard case in a person's hands, with the file already complete.